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SECTION: BSCpE 3-4	Professor: Dr. Lutzer U. Reyes
INSTRUCTIONS: • Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. • Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

Packet Tracer - Switch Security Configuration

VLAN Table

Switch	VLAN Number	VLAN Name	Port Membership	Network
SW-1	10	Admin	F0/1, F0/2	192.168.10.0/24
	20	Sales	F0/10	192.168.20.0/24
	99	Management	F0/24	192.168.99.0/24
	100	Native	G0/1, G0/2	None
	999	BlackHole	All unused	None
SW-2	10	Admin	F0/1, F0/22	192.168.10.0/24
	20	Sales	F0/10	192.168.20.0/24
	99	Management	F0/24	192.168.99.0/24
	100	Native	None	None
	999	BlackHole	All unused	None

Objectives

Part 1: Create a Secure Trunk

Part 2: Secure Unused Switchports

Part 3: Implement Port Security

Part 4: Enable DHCP Snooping

Background

You are enhancing security on two access switches in a partially configured network. You will implement the range of security measures that were covered in this module according to the requirements below. Note that routing has been configured on this network, so connectivity between hosts on different VLANs should function when completed.

Instructions

Part 1: Create a Secure Trunk.

- a. Connect the G0/2 ports of the two access layer switches.
- b. Configure ports G0/1 and G0/2 as static trunks on both switches.
- c. Disable DTP negotiation on both sides of the link.
- d. Create VLAN 100 and give it the name Native on both switches.
- e. Configure all trunk ports on both switches to use VLAN 100 as the native VLAN.

SW1 and SW2:

```
interface range GigabitEthernet0/1 - 2
switchport mode trunk
switchport nonegotiate
vlan 100
name Native
interface range GigabitEthernet0/1 – 2
switchport trunk native vlan 100
```

Part 2: Secure Unused Switchports.

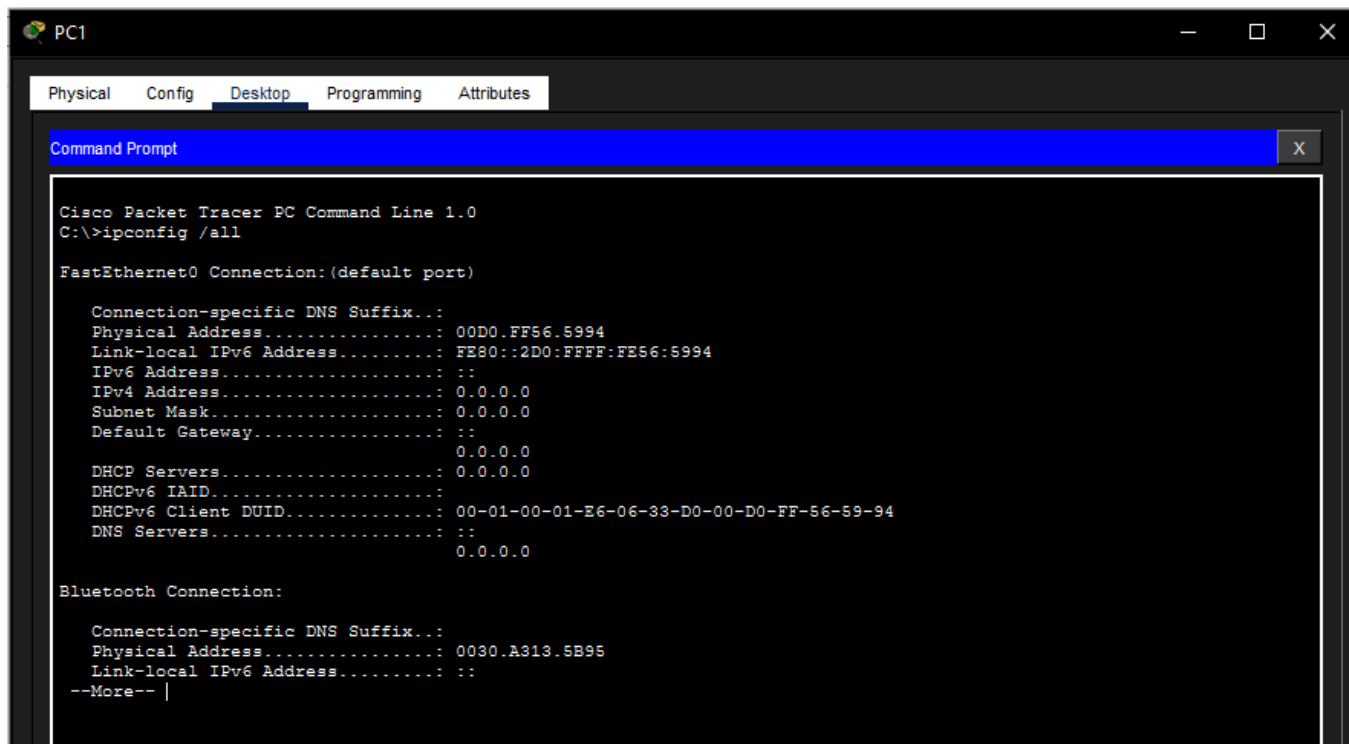
- a. Shutdown all unused switch ports on SW-1.
- a. On SW-1, create a VLAN 999 and name it BlackHole. The configured name must match the requirement exactly.
- b. Move all unused switch ports to the BlackHole VLAN.

SW1:

```
interface range FastEthernet0/3-9, FastEthernet0/11-23
shutdown
exit
vlan 999
name BlackHole
exit
interface range FastEthernet0/3-9, FastEthernet0/11-23
switchport access vlan 999
```

Part 3: Implement Port Security.

- a. Activate port security on all the active access ports on switch SW-1.
- a. Configure the active ports to allow a maximum of 4 MAC addresses to be learned on the ports.
- b. For ports F0/1 on SW-1, statically configure the MAC address of the PC using port security.
- c. Configure each active access port so that it will automatically add the MAC addresses learned on the port to the running configuration.
- d. Configure the port security violation mode to drop packets from MAC addresses that exceed the maximum, generate a Syslog entry, but not disable the ports.



The screenshot shows the PC1 Command Prompt window in Cisco Packet Tracer. The window title is "PC1" and it has tabs for Physical, Config, Desktop, Programming, and Attributes. The "Desktop" tab is active, showing a "Command Prompt" window. The command prompt displays the output of the "ipconfig /all" command, showing network configuration details for the FastEthernet0 interface. The output includes the physical address (00D0.FF56.5994), link-local IPv6 address (FE80::2D0:FFFF:FE56:5994), IPv6 address (FE80::2D0:FFFF:FE56:5994), IPv4 address (0.0.0.0), subnet mask (0.0.0.0), default gateway (0.0.0.0), DHCP servers (0.0.0.0), DHCPv6 IAID (0.0.0.0), DHCPv6 client DUID (00-01-00-01-E6-06-33-D0-00-D0-FF-56-59-94), and DNS servers (0.0.0.0). It also shows the Bluetooth connection details.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Physical Address. . . . .: 00D0.FF56.5994
    Link-local IPv6 Address . . . . .: FE80::2D0:FFFF:FE56:5994
    IPv6 Address. . . . .: FE80::2D0:FFFF:FE56:5994
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0
    DHCP Servers . . . . .: 0.0.0.0
    DHCPv6 IAID . . . . .: 0.0.0.0
    DHCPv6 Client DUID. . . . .: 00-01-00-01-E6-06-33-D0-00-D0-FF-56-59-94
    DNS Servers . . . . .: 0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address. . . . .: 0030.A313.5B95
    Link-local IPv6 Address . . . . .: 
    --More-- |
```

SW1:

```
interface range FastEthernet0/1 - 2, FastEthernet0/10, FastEthernet0/24
switchport mode access
switchport port-security
switchport port-security maximum 4
switchport port-security mac-address sticky
switchport port-security violation restrict
interface FastEthernet0/1
switchport port-security mac-address 00D0.FF56.5994
```

Part 4: Configure PortFast, and BPDU Guard.

- a. Enable PortFast on all the access ports that are in use on SW-1.
- a. Enable BPDU Guard on all the access ports that are in use on SW-1.

SW1:

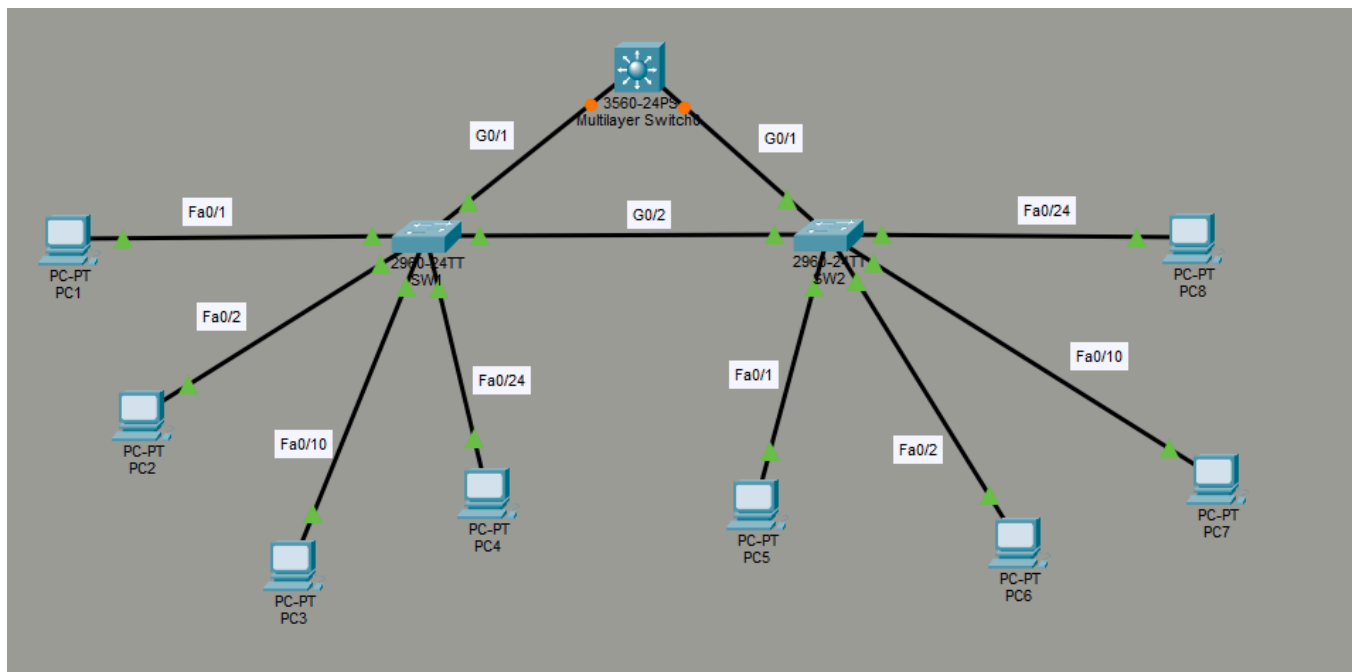
```
interface range FastEthernet0/1 - 2, FastEthernet0/10, FastEthernet0/24
spanning-tree portfast
spanning-tree bpduguard enable
```

- b. Configure SW-2 so that all access ports will use PortFast by default.

SW2:

```
spanning-tree portfast default
```

End of Document Topology and Configurations on SW1 and SW2



SW1: enable configure terminal vlan 10 name Admin vlan 20 name Sales vlan 99 name Management vlan 100 name Native vlan 999 name BlackHole exit interface range GigabitEthernet0/1 - 2 switchport mode trunk switchport nonegotiate switchport trunk native vlan 100 ip dhcp snooping trust Exit interface range FastEthernet0/3 - 9, FastEthernet0/11 - 23 switchport access vlan 999 shutdown exit interface range FastEthernet0/1 - 2, FastEthernet0/10, FastEthernet0/24 switchport mode access switchport port-security switchport port-security maximum 4 switchport port-security mac-address sticky switchport port-security violation restrict spanning-tree portfast spanning-tree bpduguard enable ip dhcp snooping limit rate 5 exit interface range FastEthernet0/1 - 2 switchport access vlan 10 exit interface FastEthernet0/10	SW2: enable configure terminal vlan 10 name Admin vlan 20 name Sales vlan 99 name Management vlan 100 name Native exit interface range GigabitEthernet0/1 - 2 switchport mode trunk switchport nonegotiate switchport trunk native vlan 100 ip dhcp snooping trust exit ip dhcp snooping ip dhcp snooping vlan 10,20,99 spanning-tree portfast default end write memory
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<pre>switchport access vlan 20 exit interface FastEthernet0/24 switchport access vlan 99 exit interface FastEthernet0/1 switchport port-security mac-address 00D0.FF56.5994 exit ip dhcp snooping ip dhcp snooping vlan 10,20,99 end write memory</pre>	
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